Al-Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Dept.

Student Name:

Microprocessor Systems
3 Credit Hours
30 NOV. 2022

Mid Exam
30 Point

Question #1: First Question (20 points): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14
B	B	B	B	A	A	A	C	A	C	D	D	D	D

Q15	Q16	Q17	Q18	Q19	Q20
B	C	C	C	C	C

1. If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?
   A. 00H to FFH B. 000H to 1FFH C. 0000H to FFFFH D. 000H to FFFH
2. What is the type of addressing mode used in instruction MOV AL, [BP+6]
   A. Indirect B. Based relative C. Direct Indexed D. Direct
3. the word length of Intel 8086 microprocessor is:
   A. 24 B. 16 C. 20 D. 32
4. 5. If the offset address of X in data segment is equal to 118H, and DS=2000H, then the physical address of X =?
   A. 21118 H B. 20118H C. 21117 H D. 31117H
5. Which is part of execution unit in Intel 8086 microprocessor?
   A. Segments registers B. ALU C. physical address generating unit D. IP register
6. The content of register BL after execution the instruction MOV BL,-7
   A. BL=F9H B. BL=FAH C. BL=FCH D. BL=FDH
7. Suppose X defined as below; what is the content of AX register after execution of the following instructions:

```
X DD 335647EBH,22H
MOV BX,OFFSET X
MOV AL,[BX]
INC BX
ADD AH,[BX+1]
```

   A. AX= 56FBH B. AX= FB56H C. X= 33FBH D. AL=47FBH
8. Which of the following flags treats with strings operations?
   A. DF B. AF C. IF D. TF
9. Find the content of AL and zero flag (ZF) and carry flag (CF) after execution of the following code:

```
MOV AL,FDH
ADD AL,3
ADC AL,3
```

   A. AL=4 ; ZF=0;CF=0 B. AL=4; ZF=0;CF=1 C. AL=2 ; ZF=0;CF=0 D. AL=2 ; ZF=1;CF=1

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10. Instruction Pointer (IP) contains offset address in _____ segment.

A. Data segment B. Stack segment C. Code segment D. Extra segment

11. If offset is a 12-bit number address, then the maximum possible location is:

A. 2 KB B. 1KB C. 1 MB D. 4 KB

12. show the binary value in register AL after execution the following instruction sequence,

```
MOV AL,11
MOV BL,15
SUB AL,BL
```

A. AL=11111111 B. AL=11111011 C. AL=11111110 D. AL=11111100

13. If the SP register contains the hexadecimal value F11FH, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP AX
PUSH CX
```

A.F11FH B. F11AH C. F11CH D. F11DH

14. Find the value of AX after execution the following assembler codes

```
MOV CX,5
MOV AX,0
L: ADD AX,CX
DEC CX
CMP CX,2
JA L
```

A. 9 B. 15 C.8 D.12

15. Number of the times the instruction sequence below will loop before coming out of loop is:

```
MOV AL,FCH
L: ADD AL,1
JNZ L
```

A.255 B. 4 C.2 D. 3

16. Assume that X memory location defined as below find the content of register DX and BX after execution of the following code:

```
X DD 13457ABCH
MOV DX,X
MOV BX,235FH
PUSH BX
PUSH DX
XCHG DL,BH
POP DX
```

A. BX= 235FH;DX=7ABCH B. BX=BC5FH ;DX= 7A23H

C .BX=BC5FH;DX=7ABCH D.BX=BC23H ;DX= 7A23H

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17. Find the content of memory location X after execution of the following code:

```
MOV AX,14
MOV BL,4
DIV BL
XOR AH,0FH
```

A. D0H B.C0H C.0DH D. 0CH

18. The binary value of AL and carry flag (CF)after the following instructions have executed is .....

```
MOV AL,10101011B
SAR AL,1
```

A AL=01010100B ; CF=0.; B.AL=01010101B;CF=1 C. AL=11010101B;CF=1 ;D.AL=01010100B ; CF=0

19. In the following instruction sequence, show the values of the Carry, and Zero flags:

```
MOV AL,00110011b
TEST AL,4
```

A. CF=1 ; ZF= 0; B. CF=0; ZF= 0 ; C. CF=0 ; ZF= 1 D.F=1 ZF= 1

20. which of the following instruction can be used to multiply constants in register DX by 8?

A. MOV CL,2; RCL DX,1; B. MOV CL,2; ROL DX,1
C. MOV CL,3; SHL DX,CL D. MOV CL,3; SAR DX,1

Question #2: (10 points)

Suppose we have a microprocessor with 32 bits address bus:

a) What is the maximum memory capacity will be?

memory capacity = 2^{32} by 1-e

b) Write the equivalent code of LOOP L

```
MOV AX,5
ADD AX,2
DEC AX
```

c) Write assembler codes to test bit#2 in register BL; if it is 1 clear bit#3 and bit#5; else set bit #4.

d) Write an assembler code to implement the following if statement :

if x< 8 s=s+15 else s=s+7; assume x and y are signed values.

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